Logarithms and Their Properties

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Evaluating logarithms**

Evaluate each logarithm exactly:

a $\log_2 32 = 5$ **b** $\log_3 \frac{1}{9} = -2$ **c** $\log 1000 = 3$ **d** $\ln e^7 = 7$

(a) $= 5$; (b) $= -2$; (c) $= 3$; (d) $= 7$

2 **[Foundational]** • **Target: Evaluating logarithms**

Use the change-of-base formula to evaluate $\log_5 18$, to three decimal places.

$\log_5 18 = \frac{\ln 18}{\ln 5} \approx \frac{2.8904}{1.6094} \approx 1.796$.

3 **[Foundational]** • **Target: Evaluating logarithms**

State the domain of $f(x) = \log(x - 4)$ and the inverse function of $g(x) = 2^x$.

Domain of f: $x > 4$. Inverse of g: $g^{-1}(x) = \log_2 x$.

4 **[Foundational]** • **Target: Properties of logarithms**

Expand fully using properties of logarithms: $\ln\left(\frac{x^3\sqrt{y}}{z^2}\right)$.

$3\ln x + \frac{1}{2}\ln y - 2\ln z$.

5 **[Foundational]** • **Target: Properties of logarithms**

Write as a single logarithm: $2\log x + 3\log y - \log z$.

$\log\left(\frac{x^2y^3}{z}\right)$.

6 **[Foundational]** • **Target: Properties of logarithms**

Given $\log_b 2 = 0.431$ and $\log_b 3 = 0.683$, evaluate:

a $\log_b 6 = 1.114$ **b** $\log_b 8 = 1.293$ **c** $\log_b 1.5 = 0.252$

(a) $= 1.114$; (b) $= 1.293$; (c) $= 0.252$

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7 [Foundational] • Target: Graphs of logarithmic functions

For $y = \log_2(x+4) - 1$, state:

a the domain $x > -4$ **b** the vertical asymptote $x = -4$ **c** the x-intercept $0 = \log_2(x+4) - 1 \Rightarrow x = -2$

(a) $x > -4$; (b) $x = -4$; (c) $0 = \log_2(x+4) - 1 \Rightarrow x = -2$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Graphs of logarithmic functions

Describe how the graph of $g(x) = -\ln(x-2) + 3$ is obtained from $y = \ln x$.

Shift right 2 units, reflect across the x-axis, then shift up 3 units.

9 [Intermediate] • Target: Logarithmic scales and applications

The Richter scale measures earthquake magnitude by $M = \log\left(\frac{I}{I_0}\right)$, where I is intensity and I_0 is a reference intensity.

a An earthquake has $M = 6$. Express I in terms of I_0. $I = I_0 \cdot 10^6$.

b How many times more intense is a magnitude-7 earthquake than a magnitude-5 earthquake? Difference in magnitude is 2, so intensity ratio is $10^2 = 100$ times.

(a) $I = I_0 \cdot 10^6$.; (b) Difference in magnitude is 2, so intensity ratio is $10^2 = 100$ times.

10 [Intermediate] • Target: Logarithmic scales and applications

Sound intensity level in decibels is $\beta = 10\log\left(\frac{I}{I_0}\right)$. If a sound's intensity doubles, by how much does β increase, to two decimal places?

$\Delta\beta = 10\log(2) \approx 10(0.3010) \approx 3.01$ dB.

11 [Intermediate] • Target: More practice with logarithm properties

Expand fully: $\log_2\left(\frac{8x^2}{\sqrt[3]{y}}\right)$.

$\log_2 8 + 2\log_2 x - \frac{1}{3}\log_2 y = 3 + 2\log_2 x - \frac{1}{3}\log_2 y$.

12 [Intermediate] • Target: More practice with logarithm properties

Write as a single logarithm and simplify: $\frac{1}{2}\ln 16 + \ln 3 - \ln 2$.

$\ln(16^{1/2}) + \ln 3 - \ln 2 = \ln 4 + \ln 3 - \ln 2 = \ln\left(\frac{4 \cdot 3}{2}\right) = \ln 6$.

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13 [Intermediate] • Target: Logarithms as inverse functions

The graphs of $y = 2^x$ and $y = \log_2 x$ are shown together with the line $y = x$. Explain why the two curves are reflections of each other across $y = x$.

Since $y = \log_2 x$ is defined as the inverse of $y = 2^x$, swapping the roles of x and y (which is exactly what reflecting across the line $y = x$ does geometrically) transforms one graph into the other.

14 [Intermediate] • Target: Solving simple logarithmic equations

Solve $\log_3 (x + 2) = 4$ for x.

$x + 2 = 3^4 = 81 \Rightarrow x = 79$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Solving simple logarithmic equations • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Solve $2\log x = \log(x + 6) + \log 1$ for x (noting $\log 1 = 0$), and check for extraneous solutions.

$2\log x = \log(x + 6) \Rightarrow \log(x^2) = \log(x + 6) \Rightarrow x^2 = x + 6 \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x - 3)(x + 2) = 0$. $x = 3$ or $x = -2$. Since $\log x$ requires $x > 0$, $x = -2$ is extraneous. Solution: $x = 3$.

16 [Challenging] • Target: The pH scale (a logarithmic application) • Cross-Lesson: 3.01 The Unit Circle and Exact Values

pH is defined as $pH = -\log[H^+]$, where $[H^+]$ is hydrogen ion concentration (mol/L). Find the pH of a solution with $[H^+] = 3.2 \times 10^{-5}$, to two decimal places.

$pH = -\log(3.2 \times 10^{-5}) \approx -(-4.4949) \approx 4.49$.

17 [Challenging] • Target: The pH scale (a logarithmic application) • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Two solutions have pH values of 3 and 6. Find the ratio of their hydrogen-ion concentrations $[H^+]$, and state which is more acidic.

$[H^+]_1 = 10^{-3}$, $[H^+]_2 = 10^{-6}$. Ratio $= \frac{10^{-3}}{10^{-6}} = 10^3 = 1000$. The pH-3 solution has 1000 times the hydrogen-ion concentration and is far more acidic (lower pH means more acidic).

18 [Challenging] • Target: Common mistakes with logarithm properties • Cross-Lesson: 3.03 Sum and Difference Formulas

Explain why $\log(x + y) \neq \log x + \log y$ in general, using a specific numerical counterexample.

Using $x = y = 10$: $\log(10 + 10) = \log 20 \approx 1.301$, but $\log 10 + \log 10 = 1 + 1 = 2$. Since $1.301 \neq 2$, the two expressions are not equal — the correct property is $\log x + \log y = \log(xy)$, not $\log(x + y)$.

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19 [Challenging] • Target: Common mistakes with logarithm properties • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Simplify $\log_2(4x) - \log_2(2x)$ using the quotient property, and verify with a specific value of x.

$\log_2(4x) - \log_2(2x) = \log_2\left(\frac{4x}{2x}\right) = \log_2(2) = 1$. Check with $x = 5$:
$\log_2(20) - \log_2(10) \approx 4.32 - 3.32 = 1$ ✓.

20 [Challenging] • Target: Solving simple logarithmic equations • Cross-Lesson: 4.01 Circles and Parabolas

Find the projection of $\mathbf{u} = \langle 6, -2 \rangle$ onto $\mathbf{v} = \langle 2, -1 \rangle$.

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|^2}\mathbf{v} = \frac{14}{5}\langle 2, -1 \rangle = \left\langle \frac{28}{5}, -\frac{14}{5} \right\rangle$. Final Answer: $\left\langle \frac{28}{5}, -\frac{14}{5} \right\rangle$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Definition of Logarithm and Inverses

Convert $\log_2(32) = 5$ to its equivalent exponential form.

a. $2^5 = 32$ ✓
b. $5^2 = 32$
c. $32^{1/5} = 2$
d. $2 \times 5 = 10$

By definition, $\log_b(y) = x$ is equivalent to $b^x = y$. Here $2^5 = 32$.

22 [Foundational] • Target: Definition of Logarithm and Inverses

Evaluate $\log_3\left(\frac{1}{27}\right)$.

a. -3 ✓
b. 3
c. $1/3$
d. $-1/3$

$3^{-3} = \frac{1}{3^3} = \frac{1}{27}$, so $\log_3(1/27) = -3$.

23 [Foundational] • Target: Definition of Logarithm and Inverses

What is the inverse function of $f(x) = e^{2x-1}$?

a. $f^{-1}(x) = \frac{\ln x + 1}{2}$ ✓
b. $f^{-1}(x) = \ln(2x - 1)$
c. $f^{-1}(x) = 2\ln x - 1$
d. $f^{-1}(x) = e^{-(2x-1)}$

$y = e^{2x-1} \Rightarrow \ln y = 2x - 1 \Rightarrow 2x = \ln y + 1 \Rightarrow x = \frac{\ln y + 1}{2}$.

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24 [Foundational] • Target: Product, Quotient, and Power Rules

Expand $\log_b\left(\frac{x^3y}{z^2}\right)$ into individual logarithms.

a. $3\log_b x + \log_b y - 2\log_b z$ ✓

b. $3\log_b x \cdot \log_b y - 2\log_b z$

c. $6(\log_b x + \log_b y - \log_b z)$

d. $\frac{3\log_b x + \log_b y}{2\log_b z}$

Apply the quotient rule, product rule, and power rule: $\log_b(x^3y) - \log_b(z^2) = 3\log_b x + \log_b y - 2\log_b z$.

25 [Foundational] • Target: Product, Quotient, and Power Rules

Condense into a single logarithm: $2\ln(x) + \frac{1}{2}\ln(y) - 3\ln(z)$.

a. $\ln\left(\frac{x^2\sqrt{y}}{z^3}\right)$ ✓

b. $\ln(x^2 + \sqrt{y} - z^3)$

c. $\ln\left(\frac{2x \cdot \frac{1}{2}y}{3z}\right)$

d. $\ln(x^2y^{1/2}z^3)$

Power rule: $\ln(x^2) + \ln(\sqrt{y}) - \ln(z^3) = \ln\left(\frac{x^2\sqrt{y}}{z^3}\right)$.

26 [Foundational] • Target: Product, Quotient, and Power Rules

Given $\log_a 2 \approx 0.301$ and $\log_a 3 \approx 0.477$, approximate $\log_a 18$.

a. 1.255 ✓

b. 0.778

c. 1.079

d. 0.954

$\log_a(18) = \log_a(2 \cdot 3^2) = \log_a 2 + 2\log_a 3 \approx 0.301 + 2(0.477) = 0.301 + 0.954 = 1.255$.

27 [Foundational] • Target: Change of Base Formula

State the change of base formula to express $\log_b a$ in terms of natural logarithms.

a. $\log_b a = \frac{\ln a}{\ln b}$ ✓

b. $\log_b a = \frac{\ln b}{\ln a}$

c. $\log_b a = \ln(a - b)$

d. $\log_b a = \ln(a) \cdot \ln(b)$

The change of base formula is $\log_b a = \frac{\log_c a}{\log_c b} = \frac{\ln a}{\ln b}$.

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28 [Foundational] • Target: Change of Base Formula

Evaluate $\log_4(32)$ using the change of base formula with base 2.

a. $5/2 = 2.5$ ✓

b. 8

c. 2

d. 5/4

$\frac{\log_2 32}{\log_2 4} = \frac{5}{2} = 2.5$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Change of Base Formula

Evaluate the product $\log_2 3 \cdot \log_3 4 \cdot \log_4 8$.

a. 3 ✓

b. 2

c. 4

d. $\log_2 8$ is 3

Using natural logs: $\frac{\ln 3}{\ln 2}\frac{\ln 4}{\ln 3}\frac{\ln 8}{\ln 4} = \frac{\ln 8}{\ln 2} = \log_2 8 = 3$.

30 [Intermediate] • Target: Domains and Graphs of Logarithmic Functions

Find the domain and vertical asymptote of $f(x) = \ln(2x - 6)$.

a. **Domain:** $(3, \infty)$**, Asymptote:** $x = 3$ ✓

b. Domain: $[3, \infty)$, Asymptote: $x = 3$

c. Domain: $(6, \infty)$, Asymptote: $x = 6$

d. Domain: $(0, \infty)$, Asymptote: $x = 0$

Argument must be strictly positive: $2x - 6 > 0 \Rightarrow 2x > 6 \Rightarrow x > 3$. Vertical asymptote at $x = 3$.

31 [Intermediate] • Target: Domains and Graphs of Logarithmic Functions

What is the range of any basic logarithmic function $f(x) = \log_b(x)$ $(b > 1)$?

a. $(-\infty, \infty)$ ✓

b. $(0, \infty)$

c. $[0, \infty)$

d. $(-\infty, 0)$

As $x \to 0^+$, $\log_b x \to -\infty$; as $x \to \infty$, $\log_b x \to \infty$. The range is all real numbers.

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32 [Intermediate] • Target: Domains and Graphs of Logarithmic Functions

Find the x-intercept of $y = \log_3(x+2) - 1$.

a. $(1, 0)$ ✓

b. $(0, 0)$

c. $(3, 0)$

d. $(-1, 0)$

Set $y = 0$: $\log_3(x+2) = 1 \Rightarrow x + 2 = 3^1 = 3 \Rightarrow x = 1$.

33 [Intermediate] • Target: Domains and Graphs of Logarithmic Functions

Expand $\log(x^3)$ using properties of logarithms.

a. $\log(x+y)$

b. $3\log x$ ✓

c. $\log(x) \cdot \log(y)$

d. $\frac{\log x}{\log y}$

Applying Exponential & Logarithmic Functions principles, Power rule: $\log(x^n) = n\log x$. Therefore, the verified correct option is $3\log x$.

34 [Intermediate] • Target: Evaluating logarithms

A 15 m rope is tied to the top of a pole and staked to level ground 9 m from the base of the pole. Find the angle the rope makes with the ground, to the nearest tenth of a degree.

a. 36.9°

b. 31.0°

c. 58.1°

d. 53.1° ✓

In the right triangle, adjacent $= 9$, hypotenuse $= 15$, so the angle is $\cos^{-1}\left(\frac{3}{5}\right) \approx 53.1^\circ$.

35 [Intermediate] • Target: Properties of logarithms

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 6)

a. $\frac{\log x}{\log y}$

b. $\log(x) \cdot \log(y)$

c. $\log x + \log y$

d. $\log x - \log y$ ✓

Quotient rule: $\log(x/y) = \log x - \log y$.

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36 [Intermediate] • Target: Graphs of logarithmic functions

Expand $\ln\left(\frac{x^3}{y}\right)$ using properties of logarithms. (variant 4)

a. $3\ln x - \ln y$ ✓

b. $\log(x) \cdot \log(y)$

c. $\frac{\log x}{\log y}$

d. $3\ln x + \ln y$

Power and quotient rules for natural log.

37 [Intermediate] • Target: Logarithmic scales and applications

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 1)

a. $\frac{\log x}{\log y}$

b. $\log x - \log y$ ✓

c. $\log(x) \cdot \log(y)$

d. $\log x + \log y$

Quotient rule: $\log(x/y) = \log x - \log y$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: More practice with logarithm properties • Cross-Lesson: 3.03 Sum and Difference Formulas

Expand $\ln(xy^2)$ using properties of logarithms. (variant 3)

a. $\log(x) \cdot \log(y)$

b. $\ln x - 2\ln y$

c. $\ln x + 2\ln y$ ✓

d. $\frac{\log x}{\log y}$

Product and power rules for natural log.

39 [Challenging] • Target: Logarithms as inverse functions • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Expand $\ln(xy^2)$ using properties of logarithms. (variant 1)

a. $\ln x - 2\ln y$

b. $\log(x) \cdot \log(y)$

c. $\frac{\log x}{\log y}$

d. $\ln x + 2\ln y$ ✓

Product and power rules for natural log.

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40 [Challenging] • Target: Solving simple logarithmic equations • Cross-Lesson: 4.01 Circles and Parabolas

Expand $\log(xy)$ using properties of logarithms. (variant 6)

a. $\log(x) \cdot \log(y)$

b. $\log x + \log y$ ✓

c. $\log x - \log y$

d. $\frac{\log x}{\log y}$

Product rule: $\log(xy) = \log x + \log y$.

41 [Challenging] • Target: The pH scale (a logarithmic application) • Cross-Lesson: 4.02 Ellipses

Expand $\log(xy)$ using properties of logarithms. (variant 4)

a. $\log x - \log y$

b. $\frac{\log x}{\log y}$

c. $\log x + \log y$ ✓

d. $\log(x) \cdot \log(y)$

Product rule: $\log(xy) = \log x + \log y$.

42 [Challenging] • Target: Common mistakes with logarithm properties • Cross-Lesson: 4.03 Hyperbolas

Expand $\log(xy)$ using properties of logarithms.

a. $\log x - \log y$

b. $\log(x) \cdot \log(y)$

c. $\frac{\log x}{\log y}$

d. $\log x + \log y$ ✓

Product rule: $\log(xy) = \log x + \log y$.

43 [Challenging] • Target: Definition of Logarithm and Inverses • Cross-Lesson: 5.01 Vectors: Components and Operations

Expand $\log(xy)$ using properties of logarithms. (variant 5)

a. $\log x - \log y$

b. $\frac{\log x}{\log y}$

c. $\log(x) \cdot \log(y)$

d. $\log x + \log y$ ✓

Product rule: $\log(xy) = \log x + \log y$.

44 [Challenging] • Target: Product, Quotient, and Power Rules • Cross-Lesson: 5.02 The Dot Product

Expand $\log\left(\frac{x^2}{y^3}\right)$ using properties of logarithms.

a. $\log(x) \cdot \log(y)$

b. $2\log x - 3\log y$ ✓

c. $2\log x + 3\log y$

d. $\frac{\log x}{\log y}$

Combine power and quotient rules.

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45 [Challenging] • Target: Change of Base Formula • Cross-Lesson: 5.03 Parametric Equations

Expand $\log\left(\frac{x^2}{y^3}\right)$ using properties of logarithms. (variant 3)

a. $\frac{\log x}{\log y}$

b. $2\log x + 3\log y$

c. $\log(x) \cdot \log(y)$

d. $2\log x - 3\log y$ ✓

Combine power and quotient rules.

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